

Propagating uncertainties in large-scale hemodynamics models via network uncertainty quantification and reduced-order modeling

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Highlights

- The NetUQ method is scalable for non-linear vector problems in realistic 3D domains.
- The TEPEM reduced-order model enables rapid and agile sampling in 3D domains.
- NetUQ+TEPEM enable cheap and accurate uncertainty quantification in large networks.
- The highly non-uniform pattern of the wall shear stress is now easily accessible.

Abstract

Numerical simulations of the cardiovascular system are affected by uncertainties arising from a substantial lack of data related to the boundary conditions and the physical parameters of the mathematical models. Quantifying the impact of this uncertainty on the numerical results along the circulatory network is challenged by the complexity of both the morphology of the domain and the local dynamics. In this paper, we propose to integrate (i) the Transverse Enriched Pipe Element Methods (TEPEM) as a reduced-order model for effectively computing the 3D local hemodynamics; and (ii) a combination of uncertainty quantification via Polynomial Chaos Expansion and classical relaxation methods – called network uncertainty quantification (NetUQ) – for effectively propagating random variables that encode uncertainties throughout the networks. The results demonstrate the computational effectiveness of computing the propagation of uncertainties in networks with nontrivial topology, including portions of the cerebral and the coronary systems.

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1. Introduction

The past several decades have witnessed a breakthrough of computational hemodynamics into medical research and clinical practice as a tool to support clinicians in the decision-making process of surgical planning and clinical

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trials [1–5]. This success has been achieved due to the maturity reached by mathematical and computational models, as well as the availability and accessibility of computational resources. However, simulating the complex physiological, biological, and mechanical dynamics of the entire cardiovascular system (CVS) or of portions of it beyond the proof-of-concept stage remains an open challenge [6], as the amount of computational power required to simulate large-scale high-fidelity models can be prohibitive, and access to off-site or remote computational platforms may be impractical for many users (e.g., hospitals) due to prohibitive costs, privacy concerns, and security policies [7–9].

Reduced-order models (ROMs) have been developed to alleviate the intrinsic complexity of high-fidelity models for hemodynamics. One-dimensional models, dating back to the work of Euler, average the cross-sectional dynamics, resulting in a 1D description of the blood vessel, representative only of the axial dynamics [5,10–13]. Then, the effect of other physiological compartments or of the peripheral circulation can be encoded via lumped-parameter models [5,12,14–16]. However, in 1D models, the transverse dynamics, which play a major role in the initiation and development of diseases like stenoses or aneurysms [17] are completely neglected. This severely restricts their reliability for clinical investigations. To address this limitation, an alternative modeling approach termed *hierarchical model reduction* (HiMod) was proposed in [18–20]; this approach improves accuracy by embedding the transverse dynamics within a reduced-order-modeling strategy. HiMod aimed at the construction of “psychologically 1D” models, where transverse details are added to the axial backbone represented by either Finite Elements or IsoGeometric Analysis with a small number of spectral modes [21–23]. In 3D this can be achieved with either a polar frame of reference [20] or with Cartesian approximate mappings [24–26]. The former tends to perform better in axisymmetric geometries, while the latter – referred to as Transversally Enriched Pipe Element Method (TEPEM) – is more appropriate for real patient-specific geometries, including bifurcations, and for this reason will be considered hereafter.

The accurate simulation of blood flow for clinical purposes is also challenged by the incomplete (sometimes absent) or inaccurate knowledge of data needed by mathematical models, like boundary conditions, physical forces, parameters, geometry, flow distribution, and material properties [14,16,27–29]. These aspects, exacerbated by other errors (e.g., discretization errors, numerical errors), generate a discrepancy between the real and the virtual scenarios, affecting the reliability of the numerical simulations. For this reason, the design of mathematical and numerical techniques to *quantify* the model uncertainty is crucial. In the cardiovascular network, this task is challenged by the complexity of the circulatory network and the complexity of the local hemodynamics. Uncertainty quantification (UQ) studies in this field have been performed, e.g., in [30–35]. In [30], sensitivity analysis is performed in 1D models of a simple bifurcation with one source of uncertainty for each branch, and of a realistic network of 37 vessels, with one random parameter for each segment. A more sophisticated analysis is carried out in [31] by studying the sensitivity of the solution to uncertain parameters and boundary conditions in a one-dimensional arterial network with 103 segments and 47 outflow boundaries. The capability of one-dimensional models of capturing the output variability of the fractional flow reserve (FFR) based on uncertain input data is validated in [32] against high-fidelity models. While these studies are realistic in terms of dimension of the stochastic problem and in the sources of uncertainty, they still rely on simplistic 1D models. Preliminary UQ studies of 3D vascular models via stochastic collocation were first performed in [35], and were applied to idealized models and to a patient-specific geometry. In [33] the distributions of boundary conditions that match the observed data are obtained via Bayesian parameter estimation and propagated in different models. The solution of the forward problem is obtained with 3D geometric multiscale simulations on HPC facilities. The more recent studies in [34] assess confidence in predictions of wall shear stress (WSS) and wall strain in 3D patient-specific geometries by developing a stochastic submodeling approach to reduce the computational cost of several multiscale solutions required by the stochastic model.

The goal of this work is to develop an effective and accurate UQ solver for cardiovascular networks. The proposed approach relies on two innovative methodological aspects. The first one corresponds to using the TEPEM approach as an accurate reduced-order model that combines the reliability of a model including transversal dynamics (so to incorporate anatomical details and important quantities like the WSS) with the efficiency of a “psychologically 1D” approach. The second novel aspect, inspired by previous work [36,37], was recently proposed in [38] and comprises an application of classical relaxation methods to propagate random variables through arbitrary networks. We demonstrate that this network uncertainty quantification (NetUQ) method, combined with a polynomial-chaos parameterization of random variables and the TEPEM reduced-order model, accomplishes the task of performing uncertainty propagation in complex large-scale vascular networks with excellent scalability and

tractable computational cost. Sections 2 and 3 present an overview of the network approach for UQ, and the TEPEM reduced-order model, respectively. In Section 4, we focus on hemodynamics applications, and present the NetUQ–TEPEM coupling for UQ. Numerical examples are reported and analyzed in Section 5, with concluding remarks drawn in Section 6.

2. The network uncertainty quantification (NetUQ) method

Decomposable systems are a common feature of many real-life applications, such as electrical power systems, gas transfer systems, or – for the interest of this work – the cardiovascular system. For such problems, applying UQ to a full-system model composed of high-fidelity subsystem models yields an extremely challenging problem, as the computational cost of complete full-system simulations is often intractable, and different physical scales pose numerical and computational constraints that can render integrating subsystem models infeasible.

The Network Uncertainty Quantification (NetUQ) method [38] enables rapid and scalable simulation of uncertainty propagation for decomposable systems by employing a bottom-up approach to perform UQ (and potentially model reduction) at the subsystem level, and by propagating uncertainties at the system level through the application of classical relaxation methods. The mathematical formulation of the method relies on modeling a decomposable system as a directed graph or networks, where each subsystem corresponds to a node, random variables exchanged between subsystems correspond to directed edges, and the system topology (i.e. the interfaces between subdomains) is encoded by the network connectivity (or adjacency matrix). More specifically, each component features input/output ports, which allow for exchanging random variables with adjacent components, or for receiving global random variables. As such, coupling (interface) conditions are formulated so that the random variables between matching coupling variables are equivalent, i.e., they provide an identical mapping from the underlying sample space to the image of the random variables.

Benefits of NetUQ include the following: (i) it requires no full-system simulations, (ii) global uncertainties can be characterized in terms of local uncertainties, and (iii) the approach is flexible and versatile due to its modularity: components can employ different uncertainty-propagation methods (e.g., stochastic collocation, stochastic Galerkin, non-intrusive spectral projection), can be characterized by different physics, and can employ different deterministic solvers.

2.1. Problem formulation

For simplicity, we present the NetUQ method as applied to random variables characterized by polynomial-chaos expansions. Consider a generic physical component i with *stochastic inputs* represented by vectors of polynomial-chaos coefficients $\mathbf{y}_i \in \mathbb{R}^{y_i}$ and $\mathbf{u}_i \in \mathbb{R}^{u_i}$, which are mapped to corresponding *stochastic outputs* represented by polynomial-chaos coefficients $\mathbf{x}_i \in \mathbb{R}^{x_i}$ via an *uncertainty-propagation operator* $\mathbf{f}_i : \mathbb{R}^{y_i} \times \mathbb{R}^{u_i} \rightarrow \mathbb{R}^{x_i}$ (e.g., provided by stochastic collocation) such that

$$\mathbf{x}_i = \mathbf{f}_i(\mathbf{y}_i, \mathbf{u}_i), \quad i = 1, \dots, n \quad (1)$$

(see Fig. 1, left). *Endogenous* inputs $\mathbf{y}_i$ correspond to outputs of adjacent components, whereas *exogenous* inputs $\mathbf{u}_i$ arise from externally specified uncertainties (e.g., uncertainties in material properties or boundary conditions). With n components, the problem can be expressed as

$$\mathbf{x} = \mathbf{f}(\mathbf{y}, \mathbf{u}). \quad (2)$$

where each term is the vectorization of the corresponding term in Eq. (1), i.e. $\mathbf{y} \in \mathbb{R}^y$, $\mathbf{x} \in \mathbb{R}^x$, and $\mathbf{u} \in \mathbb{R}^u$, with $\mathbf{y} := \sum_{i=1}^n \mathbf{y}_i$, $\mathbf{x} := \sum_{i=1}^n \mathbf{x}_i$, and $\mathbf{u} := \sum_{i=1}^n \mathbf{u}_i$ (see Fig. 1, right). Expressing endogenous inputs in terms of component outputs as $\mathbf{y} = \mathbf{I}_x^y \mathbf{x}$, where $\mathbf{I}_x^y$ is a selection operator comprising selected rows of the identity matrix and is computed from the network adjacency matrix, yields the fixed-point system

$$\mathbf{x} = \mathbf{f}(\mathbf{I}_x^y \mathbf{x}, \mathbf{u}), \quad (3)$$

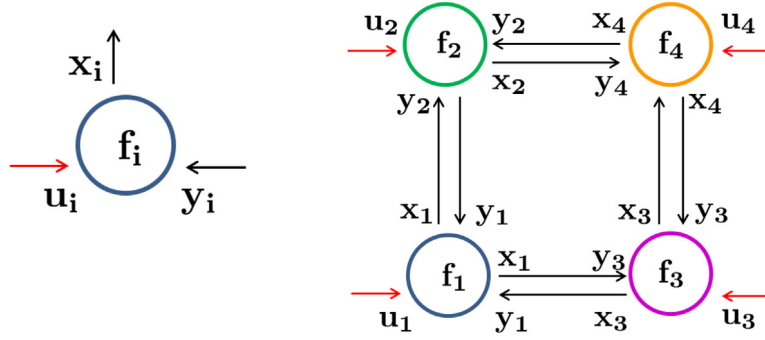


Fig. 1. Sketch of NetUQ problem formulation: single component (left) and full decomposable system (right).

or, in residual form,

$$\mathbf{x} - \mathbf{f}(\mathbf{I}_x^y \mathbf{x}, \mathbf{u}) = 0. \quad (4)$$

2.2. Uncertainty propagation

As mentioned above, we represent random variables that characterize uncertainties using polynomial chaos expansions (PCE) [39–43]. In principle, the truncation order of the input/output PCEs can be customized to input and output of each network component, based on the availability of input uncertainty information, or on the target output uncertainty. In this case, prolongation/restriction operators are necessary to match the dimensionality of the stochastic spaces across neighboring components. Such extreme flexibility in the characterization of uncertainties is a distinctive feature of NetUQ. In practice, though, in many real-life scenarios such as clinics and hospitals, input uncertainty information beyond the first order (standard deviation) is unavailable. Therefore, in the scope of this work the same truncation order is adopted for all PCEs. The more general case of customized uncertainty representations will be investigated in future work. With such model assumption, and considering the stochastic inputs $\mathbf{y}$ and $\mathbf{u}$ as functions of a multidimensional “germ” vector of independent random variables $\xi = [\xi_1, \dots, \xi_d]^T$, the corresponding PCE representations are

$$\mathbf{y}(\xi) = \sum_{k=0}^p y_k \Psi_k(\xi), \quad \mathbf{u}(\xi) = \sum_{k=0}^p u_k \Psi_k(\xi), \quad (5)$$

respectively, where $\{u_k\}_{k=0}^p, \{y_k\}_{k=0}^p$ are (deterministic) real-valued coefficients defined by a weighted L^2 -projection, and the polynomial basis functions $\{\Psi_k\}_{k \in \mathbb{N}}$ can be chosen to maximize the convergence rate according to the Askey scheme [44,45]. Notice that, in (5), all random variables throughout the network are polynomial functions of the same underlying independent germ random variables ξ . As a result, the joint probability distribution between any number of network random variables can be approximated straightforwardly as a post-processing step (e.g., by sampling the random variables and performing kernel density estimation). Each family of polynomials is orthogonal with respect to a weighting function that corresponds to the probability density function of a known random variable. For instance, assuming $\mathbf{y}$ to be a standard normal random variable, Ψ_k are Hermite polynomials that are orthogonal with respect to Gaussian density $\pi(\xi) = e^{-\xi^2/2}/\sqrt{2\pi}$. Then, the uncertainty-propagation problem consists of finding a PCE representation for the output $\mathbf{x}(\xi) = \sum_{k=0}^p x_k \Psi_k(\xi)$, as a consequence of the random input through the propagator $\mathbf{f}$, according to (2).

While PCE representations of random variables can be propagated through subsystems using a variety of uncertainty-propagation methods, this work restricts attention to *non-intrusive spectral projection* (NISP) [43], also known as discrete or pseudospectral projection. NISP interprets the PCE coefficients of the output as resulting from the orthogonal projection of the output random variable onto the PCE space such that

$$x_k = \frac{1}{\|\Psi_k\|^2} \int_{\mathbb{R}^d} \mathbf{f}(\mathbf{y}(\xi), \mathbf{u}(\xi)) \Psi_k(\xi) \pi(\xi) d\xi, \quad (6)$$

where the norm is defined as $\|\Psi_k\|^2 := \int_{\mathbb{R}^d} \Psi_k^2(\xi) \pi(\xi) d\xi$ and is known *a priori*. To make the computation of this integral tractable, NISP approximates (6) with (sparse) quadrature rules as

$$x_k \approx \frac{1}{\|\Psi_k\|^2} \sum_{q=1}^Q \tilde{f}(\mathbf{y}(\xi_q), \mathbf{u}(\xi_q)) \Psi_k(\xi_q) \pi(\xi_q) w_q, \quad (7)$$

where $\{\xi_q, w_q\}_{q=1}^Q$ are selected quadrature points and weights, respectively, and $\tilde{f} : \mathbb{R}^{\tilde{y}_i} \times \mathbb{R}^{\tilde{u}_i} \rightarrow \mathbb{R}^{\tilde{x}_i}$ is the deterministic propagator corresponding to the stochastic counterpart f . Note that, in (7), computing the term $\tilde{f}(\mathbf{y}(\xi_q), \mathbf{u}(\xi_q))$ amounts to executing one deterministic solve corresponding to the q th realization of the inputs. Therefore, differently from other intrusive approaches such as stochastic Galerkin projection, NISP does not require any modification to the underlying deterministic solver, which can be used as a black box.

2.3. Network solver

One of the most effective approaches to solve the full-system uncertainty-propagation problem (4) – which is nonlinear if the propagation operator f is nonlinear in its first argument – is Newton’s method. However, this approach requires calculating the gradient of the propagator f with respect to its first argument. Doing so is not always feasible, e.g., when NISP is employed and the deterministic solver does not provide sensitivities. Therefore, to maintain flexibility of the method, NetUQ considers the application of classical relaxation methods that do not require the gradient, promote subsystem independence, and enable acceleration methods such as Anderson Acceleration and multigrid to be employed. A block-Jacobi (i.e., additive-Schwarz) solver is equivalent to “splitting” all network edges at each iteration such that components perform uncertainty propagation independently and simultaneously. Alternatively, a block-Gauss–Seidel (i.e., multiplicative-Schwarz) solver is equivalent to cutting network edges at each iteration such that it produces a directed acyclic graph (DAG), so that each iteration corresponds to propagating random variables in a feed-forward network. While the former yields embarrassingly parallel iterations, the level of parallelization in the latter case depends on the particular DAG generated at each iteration. We refer to [38] for a rigorous formulation of the solvers (including Successive Over-Relaxation), and detailed scalability studies on benchmark problems.

3. The Transversally-Enriched Pipe-Element method (TEPEM)

Recently proposed as a numerical strategy oriented to simulate the blood flow in three-dimensional patient-specific vasculature, the TEPEM has demonstrated to be an excellent trade-off between accuracy and computational burden, which strongly limits the massive use of current numerical approximations in real clinical applications. In [25,26], and inspired in the framework of hierarchical modeling [19], this methodology was introduced and extensively tested on synthetic and patient-specific vasculatures. Several numerical results addressed in former works highlight the TEPEM ability to speed up hemodynamic simulations while maintaining a level of precision comparable to the one provided by high-fidelity FE method. The effectiveness of this novel numerical strategy relies on the combination of an *educated* discretization of the geometrical domain relying on the classical argument of the separation of variables and an improved 1D approximation for the physical fields which allows us to take into account the transverse dynamics, commonly neglected in purely 1D blood flow models. Although TEPEM was originally conceived to speed-up blood flow simulations, the strategy is quite general and can be applied to any problem formulated in pipe-like domains (e.g., gas/water pipes networks, fluvial networks, etc.).

3.1. The pipe discretization strategy

In TEPEM, the region of interest is assumed to be a connected network of tubular structures, as presented in Fig. 2. Let us consider a domain $\Omega \subset \mathbb{R}^3$ of this type. By slicing this structure along the centerline we obtain a one-dimensional partition composed by a reduced number of elements, named as pipe-elements, and generically denoted by $\mathcal{K}$. This partition, $\mathcal{T}_h(\Omega)$, is characterized by the mesh parameter h , which stands for the axial length of the pipe-elements.

Each pipe-element $\mathcal{K} \in \mathcal{T}_h(\Omega)$ clearly identifies the axial direction, where the fluid is driven, and the direction where the secondary dynamics occurs, named as *transversal direction*. For each element, a geometrical mapping

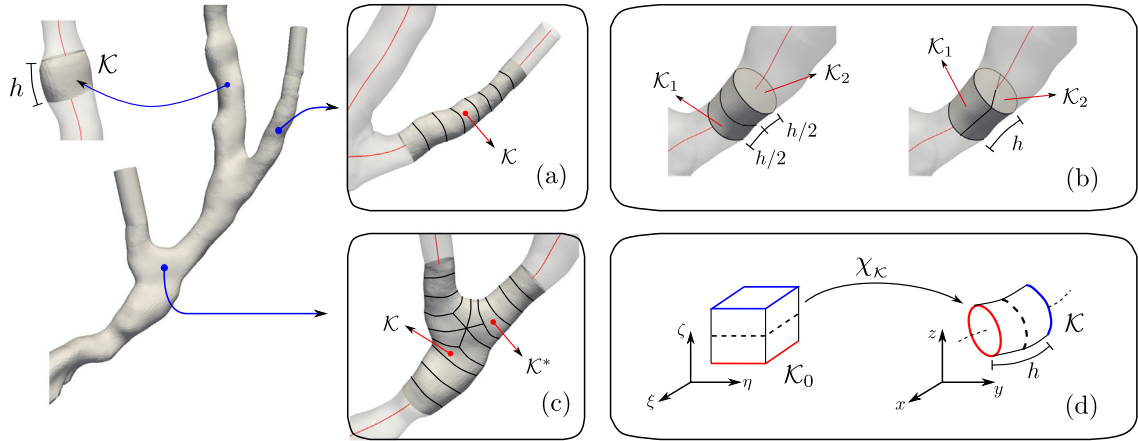


Fig. 2. Details of a pipe-type discretization of a patient-specific vasculature, red line stands by the centerline. (a) Discretization of a non-branched region slabbing the geometry along the centerline. (b) Mesh refinement by axial (left) and transversal (right) split. (c) Discretization of a junction through the use of the transition element $\mathcal{K}^*$. (d) Mapping relating deformed and reference pipe-element. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

$\chi_{\mathcal{K}} : \mathcal{K}_0 \mapsto \mathcal{K}$ is defined between the reference pipe-element $\mathcal{K}_0 = [-1, 1]^3$ (in $\xi\eta\zeta$ -space) and the physical pipe element (in xyz -space) as:

$$\chi_{\mathcal{K}}(\xi, \eta, \zeta) = \sum_{k=1}^3 \sum_{i=1}^{12} \mathbf{x}_i^k \mathcal{S}_i(\xi, \eta) \mathcal{Q}_k(\zeta), \quad (8)$$

where $\{\mathcal{S}_i : i = 1, \dots, 12\}$ is the set of cubic Serendipity functions defined in the square $[-1, 1]^2$ (see [46]), the set $\{\mathcal{Q}_k : k = 1, 2, 3\}$ is the classical Lagrangian basis for the space $\mathbb{P}_2([-1, 1])$ (quadratic polynomials defined on the segment $[-1, 1]$), and the points $\{\mathbf{x}_i^k\}$ are selected on the lateral surface of the deformed element. As a result, the geometrical representation of each pipe-element relies on the combination of piecewise cubic and quadratic order polynomials.

Pipe-type discretization in complex regions like junctions requires specific treatment for the lack of a mainstream direction inside those structures. Tubular branches are discretized by slabbing the geometry along the centerline and condensing the local secondary dynamics in a single element (Fig. 2(a)). As commonly performed in classical by-element strategies, each element can be subdivided to locally refine the mesh. In this case, each pipe-element (with length h) can be split in two ways. (i) In the axial direction, obtaining two pipe-elements with axial length $h/2$, or (ii) in the transversal direction, obtaining two pipe-elements with the same length h , each covering a portion of the transversal section. Both alternatives are outlined in Fig. 2(b). This refinement strategy allows us to naturally cluster two (or more) pipe-elements in the transversal section of regions with higher complexity (geometrical or physical) and, specifically for the case of discretized branched domains, decompose the mesh in one sub-mesh where a single pipe-element is employed for the local secondary dynamics (non-bifurcated branches) and another where two pipe-elements are considered in the transversal direction (for the discretization of the junctions). Finally, these two sub-meshes are conformally coupled through the introduction of the so-called transition element, denoted by $\mathcal{K}^*$, as can be seen in Fig. 2(c). For a deeper description of the pipe-type discretization and details of the transition element, see [26,47].

3.2. Transversally enriched approximation

The TEPEM can be understood as a smooth fashion to link one-dimensional approximations (where the transversal dynamics is entirely dropped) and full 3D approximations. The ability to *a priori* control the model capabilities relies in the way in which physical fields are approximated in the reference element $\mathcal{K}_0$. Any scalar

field w , defined in the domain Ω , is approximated by the function w^h defined by

$$w^h \circ \chi_K(\xi, \eta, \zeta) = \sum_{k=1}^{s+1} \left(\sum_{i,j=1}^{p+1} w_{ijk}^h \phi_i(\xi) \phi_j(\eta) \right) \varphi_k(\zeta), \quad K \in \mathcal{T}_h(\Omega), \quad (9)$$

where $\mathcal{T}_h(\Omega)$ is a pipe-type discretization of Ω , p and s are positive integer numbers, and the sets $\{\phi_i : i = 1, \dots, p+1\}$ and $\{\varphi_i : i = 1, \dots, s+1\}$ are Lagrangian bases for the spaces $\mathbb{P}_p([-1, 1])$ and $\mathbb{P}_s([-1, 1])$, respectively. The parameters p and s are denominated as transversal and axial polynomial order, respectively, and the functions $\{\phi_i\}_{i=1}^p$ and $\{\varphi_i\}_{i=1}^s$ as transversal and axial approximation functions. Formally, any scalar field w is approximated through a function w^h living in the finite-dimensional space $\mathbb{T}_h^{p,s}$, defined as

$$\mathbb{T}_h^{p,s} = \{w^h \in L^2(\Omega) : w^h \circ \chi_K \in [\mathbb{P}_p]^2 \times \mathbb{P}_s, K \in \mathcal{T}_h(\Omega)\}. \quad (10)$$

Moreover, thanks to the definition of the approximation functions, this structure allows to independently control the model capabilities according to the direction, enriching the capacity to represent the transverse phenomena by increasing the value of the parameter p or, analogously for the axial direction, by increasing the value of the parameter s . Our main assumption is that the local transversal dynamics on each element can be satisfactorily approached by considering a reduced number of functions, while the axial dynamics can be approximated employing low order polynomials. In a general framework, there are no specific constraints on the choice of parameters p and s , for practical purposes we combine a low degree for polynomials of the axial discretization $1 \leq s \leq 2$ (associated with an appropriate choice of the mesh size) with a moderate transversal order $4 \leq p \leq 10$. The extensive numerical experiments presented in previous works ([18,20–22] for HiMod and more specifically [25,26] for TEPEM) showed that values in those ranges are enough to reach convergence. This hybrid discretization produces “psychologically” one-dimensional finite elements with a high number of degrees of freedom (proportional to p and s); Compared to the high-fidelity 3D models, the *total* number of degrees of freedom is however highly reduced compared to conventional FEM. In this way, the computational burden is drastically reduced, yet maintaining an accurate approximation of the three-dimensional dynamics [22].

In [24], it was noticed that spurious oscillations appear in the TEPEM approximation if polynomials defined on equidistant nodes are employed as transversal functions for a Runge-like phenomenon [48]. The classical Chebyshev–Gauss–Lobatto (CGL) nodes mitigate the spurious oscillations by clustering nodes near the end-points of the interval $[-1, 1]$ (see Fig. 3) and are used instead.

4. Uncertainty quantification in blood flow problems

As illustrated in Section 1, the traditional methods that can be found in the literature for quantifying uncertainties in large-scale flow problems suffer from the intensive computational cost associated with high-fidelity numerical methods for the Navier–Stokes equations, or from the lack of transversal information intrinsic to reduced one-dimensional models. In this context, NetUQ and TEPEM feature the perfect capabilities to “fill the gap” between feasibility and reliability: while, on one hand, the NetUQ method provides an effective way to quantify the model uncertainties by promoting the independence of the subsystems, on the other hand, the TEPEM provides an effective way to solve the local problems maintaining accuracy and reduced computational burden.

Thus, in this work we embed the TEPEM strategy into the NetUQ framework, in order to provide relevant statistical information about quantities of clinical interest, such as the wall shear stress, with a level of detail currently out of reach for the traditional one-dimensional models, and in a fraction of the computational time demanded by full three-dimensional approaches.

4.1. Blood flow problem

The blood motion is described by the incompressible Navier–Stokes equations, completed with suitable boundary conditions [16]. Let us denote by $\Omega \in \mathbb{R}^3$ an isolated region from the cardiovascular system and by Γ its boundary. The whole boundary can be subdivided into three regions: inlet (Γ_i), outlet (Γ_o) and lateral (Γ_L) boundaries. We assume inlet and outlet boundaries to be planar sections, and Γ_L to be a smooth surface (see Fig. 4). At either boundaries Γ_i and Γ_o , Neumann or Dirichlet boundary conditions can be prescribed. In more realistic scenarios, we may expect that not all the data needed by the mathematical model is available (defective data problems).

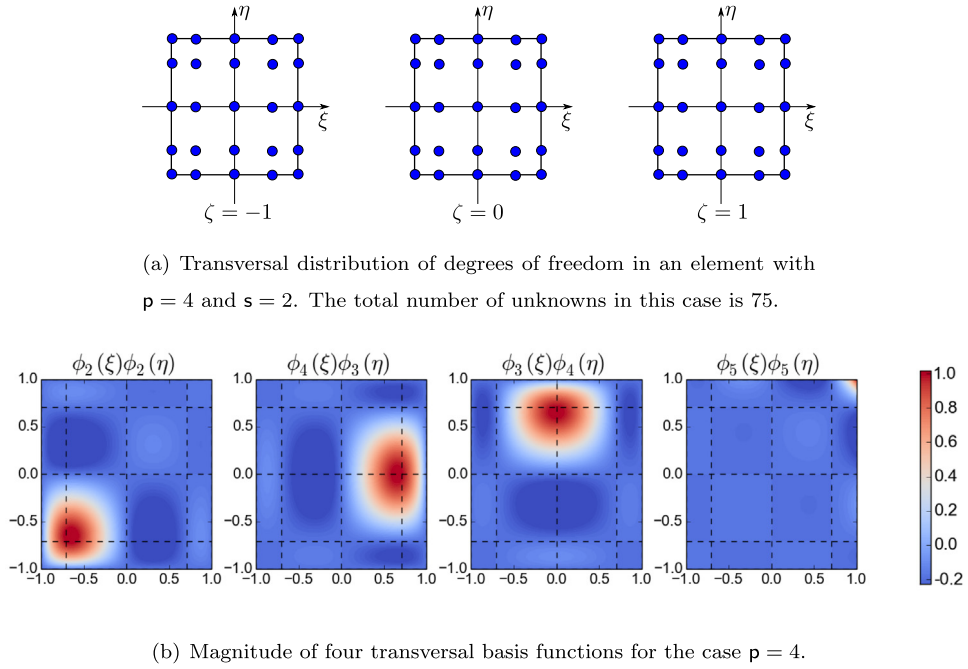


Fig. 3. (a) Geometrical distribution of the degrees of freedom, on a generic pipe-element, for the combination $p = 4$ and $s = 2$. (b) Planar view of some functions on the transversal basis for $p = 4$.

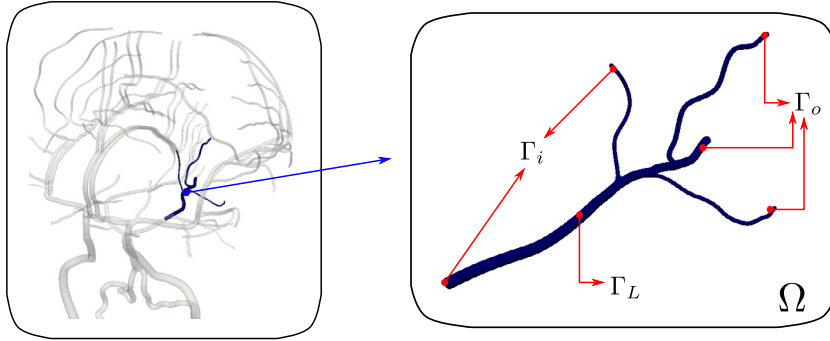


Fig. 4. Schematic setting for the model problem. In the example, the domain Ω is an isolated region from the intracranial system of the ADAN model [55].

In this case, we may consider other approaches for the prescription of boundary conditions, like flow through Lagrange multipliers [27,49,50], the distribution of a total resistance among multiple outlets [28,29], the control approach [51,52], or generally the geometric multiscale approach [12,53,54]. Over the lateral boundary Γ_L , no-slip boundary conditions are considered for the velocity field, as the vessel is assumed to be rigid.

The fluid dynamics is described through the velocity ($\mathbf{u}$) and pressure (p) fields, and the variational formulation of this problem reads as follows: find $(\mathbf{u}, p) \in \mathcal{V} \times L^2(\Omega)$ such that

$$\int_{\Omega} \left[\rho \frac{\partial \mathbf{u}}{\partial t} \cdot \hat{\mathbf{u}} + \rho (\nabla \mathbf{u}) \mathbf{u} \cdot \hat{\mathbf{u}} + 2\mu \boldsymbol{\varepsilon}(\mathbf{u}) \cdot \boldsymbol{\varepsilon}(\hat{\mathbf{u}}) - p \operatorname{div} \hat{\mathbf{u}} - \hat{p} \operatorname{div} \mathbf{u} \right] d\Omega = \int_{\Gamma_i} t_i \mathbf{n} \cdot \hat{\mathbf{u}} d\Gamma_i + \int_{\Gamma_o} t_o \mathbf{n} \cdot \hat{\mathbf{u}} d\Gamma_o \quad \forall (\hat{\mathbf{u}}, \hat{p}) \in \mathcal{V} \times L^2(\Omega), \quad (11)$$

where ρ and μ are the fluid density and viscosity, respectively, $\boldsymbol{\varepsilon}(\cdot) = (\nabla(\cdot))^S$ is the symmetric part of the gradient operator, $\mathbf{n}$ is the outward normal unit vector, the hat sign $\hat{(\cdot)}$ denotes an admissible variation of field $(\cdot)$, and t_i and t_o are given data which stand for the magnitude of the normal component of the traction vector imposed at Γ_i and Γ_o , respectively. The space $\mathcal{V}$ is defined as

$$\mathcal{V} = \{\mathbf{u} \in \mathbf{H}^1(\Omega); \mathbf{u}|_{\Gamma_L} = \mathbf{0}, (\Pi_{\mathbf{n}}\mathbf{u})|_{\Gamma_i} = \mathbf{0}, (\Pi_{\mathbf{n}}\mathbf{u})|_{\Gamma_o} = \mathbf{0}\}, \quad (12)$$

with $\mathbf{H}^1(\Omega) = [H^1(\Omega)]^3$, and where $\Pi_{\mathbf{n}} = \mathbf{I} - \mathbf{n} \otimes \mathbf{n}$ is the projection operator over the surface with normal unit vector $\mathbf{n}$. Since the domain is an isolated region of the cardiovascular system, we assume that the incoming and outgoing velocity profile is fully developed by enforcing the condition $(\Pi_{\mathbf{n}}\mathbf{u}) = \mathbf{0}$ on the inlet and outlet boundaries.

In the TEPEM scope, the numerical solution of the fluid flow problem is obtained by discretizing the domain Ω with a pipe-type mesh $\mathcal{T}_h(\Omega)$, and by approximating the velocity and pressure fields by the functions $\mathbf{u}^h \in \mathcal{V}_h^{\mathbf{p}_u, \mathbf{s}_u}$ and $\mathbf{p}^h \in \mathcal{Q}_h^{\mathbf{p}_p, \mathbf{s}_p}$, respectively, in the discrete spaces defined as

$$\mathcal{V}_h^{\mathbf{p}_u, \mathbf{s}_u} = [\mathbb{T}_h^{\mathbf{p}_u, \mathbf{s}_u}]^3 \cap C(\overline{\Omega}) \cap \mathcal{V}, \quad \mathcal{Q}_h^{\mathbf{p}_p, \mathbf{s}_p} = \mathbb{T}_h^{\mathbf{p}_p, \mathbf{s}_p} \cap C^*(\overline{\Omega}), \quad (13)$$

where $\mathbf{p}_u, \mathbf{p}_p, \mathbf{s}_u, \mathbf{s}_p \in \mathbb{N}$ are the transversal and axial orders for velocity and pressure, respectively, as discussed in Section 3, and $C^*(\overline{\Omega})$ stands for the space of continuous functions with discontinuities at the interfaces of transition elements. Inspired by classical inf–sup stable spaces for FEM and spectral methods, we set for the axial polynomial order $\mathbf{s}_u = 2$ and $\mathbf{s}_p = 1$ (1D Taylor–Hood finite elements) while, for the transversal dynamics, we set the relation $\mathbf{p}_p = \mathbf{p}_u/2$ (with $\mathbf{p}_u$ even). This combination is expected to fulfill the inf–sup condition due to the absence of spurious modes in all the simulations addressed so far. Furthermore, this combination satisfies the discrete inf–sup test proposed by Chapelle in [56], which can be understood as a numerical proof of the pair stability (see [47]). In general, the inf–sup condition for this type of hybrid-order discretizations is an open problem.

4.2. Geometrical decomposition of the vasculature

A simple procedure to discretize patient-specific vasculatures in pipe-type elements was presented in [26]. This procedure can be easily adapted to perform an automatic decomposition of the whole vasculature with overlapping components, taking each bifurcation in the geometry as a basic component of this subdivision. Then, each component is discretized with a pipe-type mesh. The decomposition process, outlined in Fig. 5, consists in the following steps:

- (i) The centerline is partitioned in overlapping single bifurcations. Since the decomposition of a one-dimensional structure is a trivial process, this step does not introduce further complexities in the whole algorithm.
- (ii) A global decomposition is performed by applying the algorithm presented in [26] to each centerline segment, to construct the vasculature components and the corresponding pipe-discretization. In each component Ω_k appear, over their boundary $\partial\Omega_k$, artificial interfaces (denoted as γ_{ik} - internal boundary of component Ω_k located inside component Ω_i) and where suitable boundary conditions need to be imposed to achieve (at the convergence) the continuity of the global velocity field.
- (iii) An auxiliary internal decomposition within each three-dimensional component is performed by splitting the overlapping section from the rest of the whole component, which allows exchanging information between neighboring subdomains. As we detail in the next subsection, each subdomain is connected to its neighbors by a Lagrange-multiplier approach that enforces the continuity of the velocity field on corresponding interface sections of adjacent components. For example, the continuity of the velocity field over the internal cross-section of Ω_2 (that corresponds to the artificial boundary of component Ω_1) is enforced through an outer traction that plays the role of a Lagrange multiplier. It is important to highlight that this subdivision is performed at the level of the pipe discretization of each component, without the creation of any additional component for the domain decomposition (DD) algorithm.

This algorithm exploits the pipe structure of the domain of interest to reduce the complex task of decomposing a three-dimensional geometry and meshing the generated overlapping components into a simple routine of decomposing a one-dimensional centerline. This 1D-like decomposition is extremely simple and inexpensive. As a matter of fact, the geometrical decomposition and meshing process for the larger geometry addressed in this work (Section 5) takes less than 0.5 min running in a workstation with a single core.

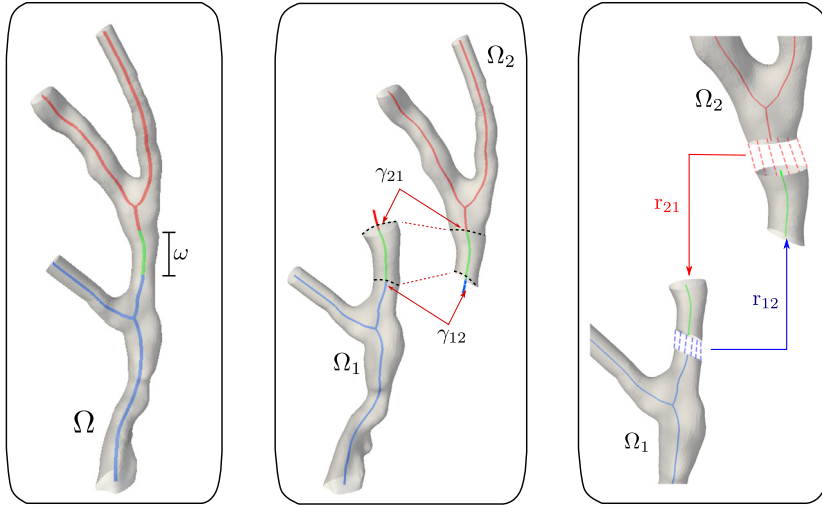


Fig. 5. Details in the geometrical decomposition. Left: Decomposition of the centerline in two components (red and blue lines) with a common shared portion (green line). Center: Generation of the three-dimensional components with the creation of interfaces (γ_{12} and γ_{21}). Right: Local decomposition within each component, and transfer of internal boundary data. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4.3. Domain decomposition formulation

In order to formulate the problem in a domain decomposition (DD) framework, we ease the notation in the definition of the blood flow problem by introducing the forms

$$\mathcal{N}_{\Omega}((\mathbf{u}, p); (\hat{\mathbf{u}}, \hat{p})) = \int_{\Omega} \left[\rho \frac{\partial \mathbf{u}}{\partial t} \cdot \hat{\mathbf{u}} + \rho (\nabla \mathbf{u}) \mathbf{u} \cdot \hat{\mathbf{u}} + 2\mu \boldsymbol{\varepsilon}(\mathbf{u}) \cdot \boldsymbol{\varepsilon}(\hat{\mathbf{u}}) - p \operatorname{div} \hat{\mathbf{u}} - \hat{p} \operatorname{div} \mathbf{u} \right] d\Omega, \quad (14)$$

$$\mathcal{L}_{\Omega}((\hat{\mathbf{u}}, \hat{p})) = \int_{\partial\Omega} \mathbf{t} \cdot \hat{\mathbf{u}} d\Gamma. \quad (15)$$

Then, the monolithic problem can be formulated as: find $(\mathbf{u}, p) \in \mathcal{V} \times L^2(\Omega)$ such that

$$\mathcal{N}_{\Omega}((\mathbf{u}, p); (\hat{\mathbf{u}}, \hat{p})) = \mathcal{L}_{\Omega}((\hat{\mathbf{u}}, \hat{p})) \quad \forall (\hat{\mathbf{u}}, \hat{p}) \in \mathcal{V} \times L^2(\Omega). \quad (16)$$

The domain decomposition problem is formulated on an overlapping partition $\{\Omega_1, \dots, \Omega_n\}$ of the region of interest Ω , i.e., $\Omega = \bigcup_{k=1}^n \Omega_k$, with $\Omega_k \subset \Omega$ and $\Omega_i \cap \Omega_j \neq \emptyset$ for any $i, j \in \{1, \dots, n\}$, $i \neq j$. For each component Ω_k , we define the set of its neighbors as $\mathcal{I}_k = \{j \in \{1, \dots, n\}, j \neq k, \text{ s.t. } \Omega_k \cap \Omega_j \neq \emptyset\}$, with cardinality $\mathbf{c}_k$. Then, using the notation (14)–(15), the additive-Schwarz DD formulation reads:

At each iteration $m = 1, 2, \dots$ until convergence, find $(\mathbf{u}_k^m, p_k^m, \mathbf{r}_k^m) \in \mathcal{V}_k \times L^2(\Omega_k) \times \mathcal{T}_k$ in each component Ω_k ($k = 1, \dots, n$), such that

$$\begin{aligned} \mathcal{N}_{\Omega_k}((\mathbf{u}_k^m, p_k^m); (\hat{\mathbf{u}}_k, \hat{p}_k)) + \sum_{i \in \mathcal{I}_k} \int_{\gamma_{ki}} \left[[\mathbf{u}_k^m] \cdot \hat{\mathbf{r}}_{ki} + [\hat{\mathbf{u}}_k] \cdot \mathbf{r}_{ki}^m \right] d\Gamma = \\ \mathcal{L}_{\Omega_k}((\hat{\mathbf{u}}_k, \hat{p}_k)) + \sum_{i \in \mathcal{I}_k} \int_{\gamma_{ik}} \mathbf{r}_{ik}^{m-1} \cdot \hat{\mathbf{u}}_k d\Gamma \quad \forall (\hat{\mathbf{u}}_k, \hat{p}_k, \hat{\mathbf{r}}_k) \in \mathcal{V}_k \times L^2(\Omega_k) \times \mathcal{T}_k, \end{aligned} \quad (17)$$

with $\hat{\mathbf{r}}_k = (\hat{\mathbf{r}}_{k1}, \dots, \hat{\mathbf{r}}_{kc_k})$, $\mathbf{r}_{ik}^0 = \mathbf{0}$ for each $i \in \mathcal{I}_k$, and the spaces $\mathcal{V}_k$ and $\mathcal{T}_k$ defined by

$$\mathcal{V}_k = \left\{ \mathbf{u} \in \mathbf{H}^1(\Omega_k); \mathbf{u}|_{\Gamma_L \cap \partial\Omega_k} = \mathbf{0}, (\boldsymbol{\Pi}_n \mathbf{u})|_{\Gamma_i \cap \partial\Omega_k} = \mathbf{0}, (\boldsymbol{\Pi}_n \mathbf{u})|_{\Gamma_o \cap \partial\Omega_k} = \mathbf{0} \right\}, \quad (18)$$

$$\mathcal{T}_k = \left\{ \mathbf{r}_k = (\mathbf{r}_{k1}, \dots, \mathbf{r}_{kc_k}) : \mathbf{r}_{ki} \in \mathbf{H}^{1/2}(\gamma_{ki}), i \in \mathcal{I}_k \right\}. \quad (19)$$

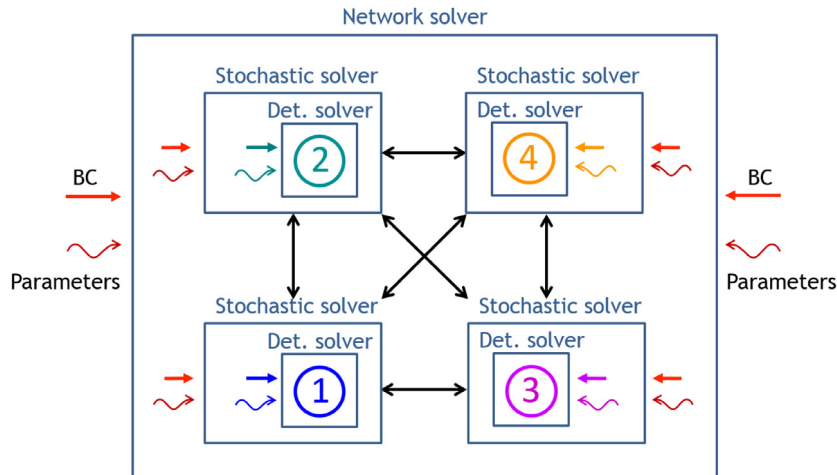


Fig. 6. Sketch of the NetUQ network solver. Straight and wavy arrows represent exogenous stochastic boundary conditions (BC) and parameters, respectively.

In each component Ω_k ($k = 1, \dots, n$), the traction $\mathbf{r}_{ki}$ is originated from the enforcement of the continuity condition for the velocity field, that is the enforcement of $[\![\mathbf{u}]\!] = \mathbf{0}$ (null jump condition), at the shared internal boundary corresponding to the adjacent domain Ω_i . Notice that, imposing the traction on each artificial boundary, we are ensuring that each local problem is well-defined with the corresponding spaces for velocity and pressure. Also, the space $\mathcal{T}_k$ is formed by the Lagrange multipliers employed to enforce the continuity of the velocity field at all internal boundaries shared between the different components. From a computational viewpoint, since we are using an additive DD formulation, the local problems are completely decoupled and can be solved simultaneously. We refer to Section 5 for details on the convergence criterion.

4.4. The NetUQ–TEPEM algorithm

Our approach is characterized by a nested structure (see Fig. 6). At the outer level, the network solver receives global stochastic BC and model parameters as inputs, and handles the relaxation iterative process until convergence. At each iteration, the exogenous inputs specific to each component are selected, and the endogenous inputs are exchanged. With a block-Jacobi network solver, since all the components are decoupled, the endogenous interface conditions (that is the traction at internal boundaries) can be exchanged simultaneously; with a Gauss–Seidel or SOR network solver, the order in which they are exchanged depends on the permutation of the network components, which can be selected to maximize the level of concurrency. Then, for each component, the local stochastic solver evaluates the Q realizations of the stochastic inputs, and calls the TEPEM deterministic solver for each instance of the inputs. Finally, the output PC coefficients are computed according to (7), and the termination criterion is checked by the network solver. The process is repeated until convergence.

The stochastic component of the problem is handled via the C++ library UQTK [57], developed at the Sandia National Laboratories, which has been embedded in a MATLAB implementation of the NetUQ network solver.

5. Numerical results

To study the numerical and predictive capabilities of the coupling NetUQ–TEPEM for the quantification of uncertainties in fluid flow simulations, two sets of studies are addressed: (i) weak scalability tests in phantom branched domains, and (ii) accuracy tests in more realistic vasculatures generated from a patient-specific medical image, and from the one-dimensional ADAN model [55]. The scalability properties of the NetUQ method have been previously tested in [38] on linear and non-linear scalar equations in 1D or 2D elementary domains. The numerical results featured excellent scaling for all the variations of the network solvers. Here, for the first time, we extend the scalability benchmarking of the NetUQ solver to *non-linear vector* problems in 3D domains. The performance of the

solver with such setting is not easily predictable, due to the complexity and intrinsic randomness of the underlying physics induced by the more complex governing equations. In addition, while the stochastic solver in [38] relies on an underlying high-fidelity FE deterministic model, here we attempt accurate UQ based on deterministic ROM, which alleviates the cost of subsystem sampling. Finally, we move from academic proof of concept to real-life applications by simulating UQ in patient-specific geometries.

5.1. Test setting

For all the cases addressed, the fluid flow is simulated in a steady state regime with deterministic fluid density $\rho = 1.04 \text{ g/cm}^3$ and viscosity $\mu = 0.04 \text{ P}$ (although stochastic model parameters are also supported by the method), and stochastic inflow/outflow boundary conditions. It is worth stressing that uncertainty in boundary data for cardiovascular simulations is an important topic, in view of massive use of numerical modeling in the clinical routine, as done, for instance, by the company HeartFlow [58]. Since a stochastic solution in closed form is unavailable, statistical moments of the quantities of interest (QoI) are computed in the whole domain monolithically as a reference solution. The random variables characterizing the system's uncertainties are assumed to be Gaussian, with variability set to 10% of the corresponding average value, and are represented via first-order PCE. The additive-Schwarz version of the NetUQ network solver with Anderson Acceleration [59] (depth $m = 6$), with termination criterion based on a relative-residual tolerance of $\varepsilon = 10^{-4}$, and the normalization factor is the norm of the first NetUQ Jacobi iterate corresponding to an initial guess of zeros for the endogenous inputs. Preliminary pilot simulations were run to determine the minimal sparse quadrature level to capture the variability of the physics ($l = 1$ for scalability studies, $l = 2$ for simulations in realistic geometries).

The same procedure was performed to reach the minimal value for transversal polynomial order for the TEPEM to reach the convergence of the Navier–Stokes equations. In all the examples, deterministic samples are solved using TEPEM with $p_u = 4$. An automatic selection of the transversal order based on *a-posteriori* error estimates has been investigated in [22,60], for 2D parabolic equations, with a combination of nodal (axial direction) and modal (transverse direction) adaptivity in the HiMod framework with Fourier spectral approximation. A similar study for fluid flow problems solved with TEPEM will be addressed in future works.

Since the focus of this preliminary work is in showing the effectiveness of the strategy, rather than quantifying the potential computational saving, the TEPEM inner solver uses a direct serial algebraic LU solver on each local problem. The natural parallelism of the system matrix assembling phase will be exploited in future studies. In contrast, due to the large size of the monolithic domain, the reference solution sampling is performed by TEPEM where the algebraic system is solved by using an iterative GMRES method. Within each subsystem (in a NetUQ or monolithic setting) the sampling is sequential, although the use of parallel computing paradigms can be applied and will be subject of future studies.

All the numerical experiments presented here have been performed on Intel(R) Core(TM) i7-2600 CPU @ 3.40 GHz machines.

5.2. Scalability tests

Weak scalability is tested on three different network configurations. Networks 1 and 2 feature a binary structure that mimics a structured branching topology of the human vascular system, with one and two global inflows, respectively. Network 3 is characterized by a denser connectivity, which allows for modeling multiple internal merging flows. In each network, the basic component is a bifurcated vessel with constant radius, where each branch may feature a different length or radius, depending on the network configuration. The description of these geometries is outlined in Fig. 7. To study how the approach scales with respect to the problem size, each network is subdivided in four levels or subnetworks.

Specifically – as mentioned previously – we consider stochastic boundary conditions and deterministic density and viscosity, although stochastic model parameters are also supported by the method. In particular, since the number of outlets increases with the network size, uncertainty is localized only at the inlets, so that the dimension of the stochastic space is the same for the different subnetworks.

The spatial discretization considers 50 pipe-elements on each basic component, corresponding to an axial discretization with length $h = 0.1 \text{ cm}$, and the overlap between components is composed of 6 pipe-elements.

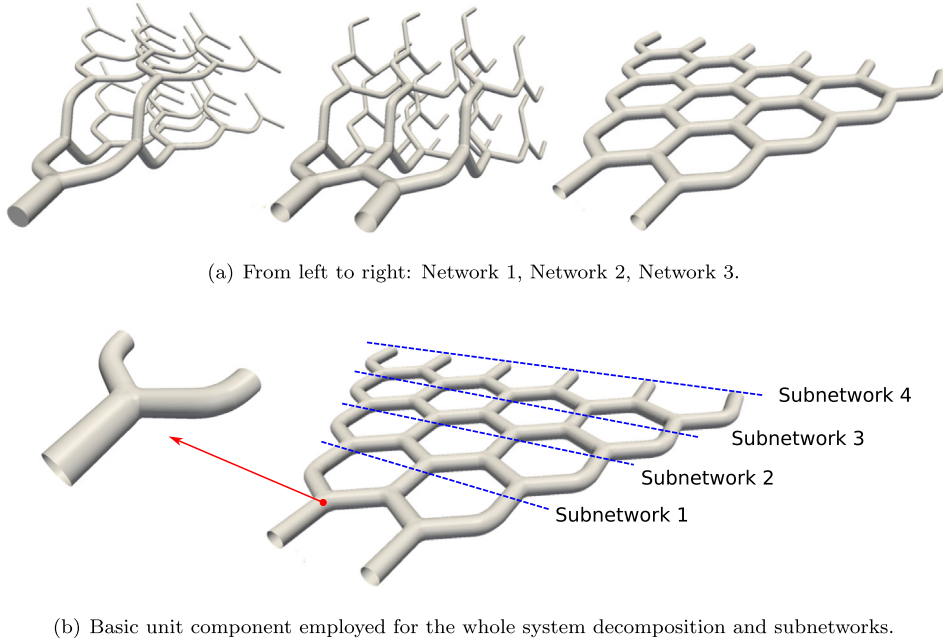


Fig. 7. Geometrical setting for the weak scalability test. Three phantom branched geometries are considered and decomposed employing a single bifurcation as basis component.

For each network configuration, the cost of a local stochastic solve is fixed as the network size increases. The scalability of the solver is measured in terms of the number of iterations to convergence and ideal parallel execution time as a function of the network size. In an ideal scenario with no constraints on computational resources, the additive network solver is embarrassingly parallelizable by defining a task as one local stochastic solve, and by creating a one-to-one task-processor mapping. Consequently, the ideal parallel time T_p (i.e., not including overhead) is computed as the sum of the execution time of the slowest processor at each iteration, over all iterations, i.e.,

$$T_p = \sum_{j=1}^{n_\varepsilon} \max_{i=1, \dots, n} T_i^j, \quad (20)$$

where n_ε is the number of iterations to convergence, n is the number of network components, and T_i^j is the execution time of component i at iteration j . Then, the ideal speedup is given by $S = T_s / T_p$, where T_s is the serial execution time, defined as the execution time to solve the stochastic problem in the corresponding monolithic network with one processor (without domain decomposition).

Fig. 8 (top) shows that the number of iterations and the ideal parallel time grow sub-linearly with respect to the number of network components, which highlights the scalability properties of the NetUQ approach also for vector problems in 3D geometries. The speedup (see Fig. 8, top-right) – roughly linear as the network size increases – shows that NetUQ outperforms traditional UQ for large-scale problems. In fact, since the *global* problem size of small networks (e.g., $n = 3$) is comparable to the *local* problem size, a monolithic approach is more computationally efficient (convergence is reached in one iteration, roughly at the same cost of one local solution). On the other hand, for larger networks, since the *local* computational burden is a *small* fraction of the *global* problem size, one stochastic monolithic solution is less efficient than one (iterative) NetUQ solution, which can count on a more effective and agile parallel execution thanks to the underlying domain decomposition approach.

The accuracy of the NetUQ solver is tested against the solution obtained with a traditional UQ approach applied to the full problem in the entire domain. Quantities such as the relative error of mean value and standard deviation normalized by the full-system solution are bounded as the network size increases (see Fig. 8, bottom). These results extend the numerical validation of the truncation error boundedness from non-linear 2D scalar problems (assessed in [38]) to 3D non-linear vector problems (3D incompressible Navier–Stokes equations). Moreover, they provide

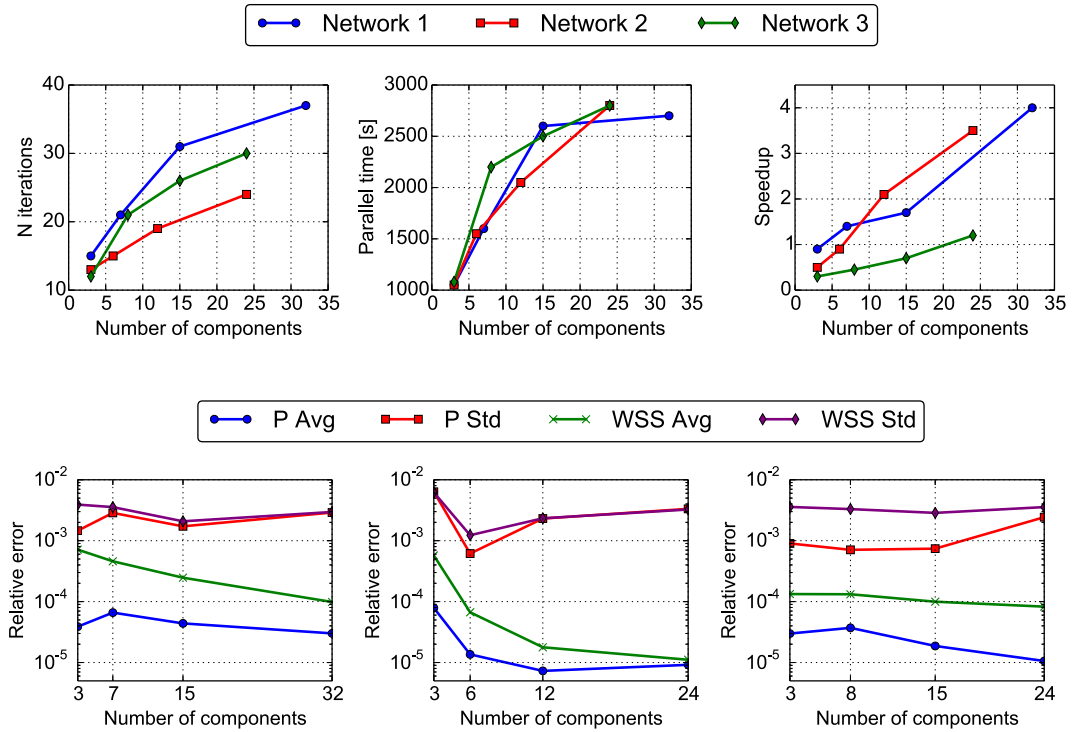


Fig. 8. Top: Number of iterations (left), ideal parallel time (center), speedup (right) of network 1 (○), network 2 (□), network 3 (◇); Bottom: relative error for pressure expected value (○) and standard deviation (□), and WSS expected value (×) and standard deviation (◇) for network 1 (left), network 2 (center) and network 3 (right).

numerical evidence of the analytical a-priori and a-posteriori error bounds derived in [38], confirming that the component PCE truncation errors do not amplify throughout the network. A detailed report of the accuracy of the proposed strategy with corresponding number of iterations to convergence, together with the total number of components and pipe-elements needed to discretize each subnetwork, is presented in Table 1. Similarly to the NetUQ scalability studies on 2D scalar problems presented in [38], the error in higher-order moments (i.e., in this case, the standard deviation) is higher than the corresponding error on lower-order moments. Moreover, the error of the expected value decreases with the network size. This is due to the fact that, in a domain-decomposition framework, adding components entails a more informative exchange of data through the interfaces, benefiting from a denser connectivity (and, consequently, communication) among components, and thus leading to a more accurate approximation. This corrective effect does not happen for higher-order moments, since the error is dominated by the PCE truncation error.

5.3. Toward realistic geometries

To further study the capabilities of the proposed approach, we increase the geometrical complexity of the domain of analysis. Specifically, we study the accuracy of the proposed strategy in two cases, named Geometry A (based on the skeleton of ADAN model and radii information) and Geometry B (based on patient-specific data). These two cases represent different levels of complexity: (i) in the number of sources of uncertainties, and (ii) in the geometrical structure (and consequently on the network size), see Fig. 9. To measure the accuracy of the coupling NetUQ–TEPEM we compare again the predicted results against a reference solution obtained with traditional UQ approach applied to the whole domain. Besides the average value and standard deviation of the quantities of interest, we are also interested in the spatial distribution of the *coefficient of variation* (CV) of the QoI, also known as *relative standard deviation* [61]. The CV is computed as the ratio of the standard deviation to the average value, and provides

Table 1

Number of components (DD) and pipe-elements (Pipe) employed to discretize each subnetwork. Also, it is presented the number of iterations to convergence, the relative error in the average (avg) and in the standard deviation (std) for the pressure and wall shear stress between the NetUQ-TEPEM solution and the full-system solution for the three networks considered on the scalability test.

		Elements		Iterations	Pressure error		WSS error	
		DD	Pipe		Avg	Std	Avg	Std
Network 1	Subnetwork 1	3	196	15	$3.88 \cdot 10^{-5}$	$1.46 \cdot 10^{-3}$	$7.14 \cdot 10^{-4}$	$3.91 \cdot 10^{-3}$
	Subnetwork 2	7	448	21	$6.60 \cdot 10^{-5}$	$2.86 \cdot 10^{-3}$	$4.59 \cdot 10^{-4}$	$3.56 \cdot 10^{-3}$
	Subnetwork 3	15	952	31	$4.39 \cdot 10^{-5}$	$1.72 \cdot 10^{-3}$	$2.47 \cdot 10^{-4}$	$2.08 \cdot 10^{-3}$
	Subnetwork 4	32	1960	37	$3.01 \cdot 10^{-5}$	$2.90 \cdot 10^{-3}$	$9.86 \cdot 10^{-5}$	$2.96 \cdot 10^{-3}$
Network 2	Subnetwork 1	3	195	13	$7.90 \cdot 10^{-5}$	$6.35 \cdot 10^{-3}$	$5.73 \cdot 10^{-4}$	$6.20 \cdot 10^{-3}$
	Subnetwork 2	6	402	15	$1.36 \cdot 10^{-5}$	$6.15 \cdot 10^{-4}$	$6.68 \cdot 10^{-5}$	$1.23 \cdot 10^{-3}$
	Subnetwork 3	12	810	19	$7.29 \cdot 10^{-6}$	$2.31 \cdot 10^{-3}$	$1.78 \cdot 10^{-5}$	$2.33 \cdot 10^{-3}$
	Subnetwork 4	24	1626	24	$9.17 \cdot 10^{-6}$	$3.33 \cdot 10^{-3}$	$1.11 \cdot 10^{-5}$	$3.25 \cdot 10^{-3}$
Network 3	Subnetwork 1	3	196	12	$2.99 \cdot 10^{-5}$	$9.01 \cdot 10^{-4}$	$1.33 \cdot 10^{-4}$	$3.58 \cdot 10^{-3}$
	Subnetwork 2	8	462	21	$3.71 \cdot 10^{-5}$	$7.08 \cdot 10^{-4}$	$1.32 \cdot 10^{-4}$	$3.30 \cdot 10^{-3}$
	Subnetwork 3	15	827	26	$1.87 \cdot 10^{-5}$	$7.40 \cdot 10^{-4}$	$9.97 \cdot 10^{-5}$	$2.86 \cdot 10^{-3}$
	Subnetwork 4	24	1291	30	$1.06 \cdot 10^{-5}$	$2.43 \cdot 10^{-3}$	$8.27 \cdot 10^{-5}$	$3.56 \cdot 10^{-3}$

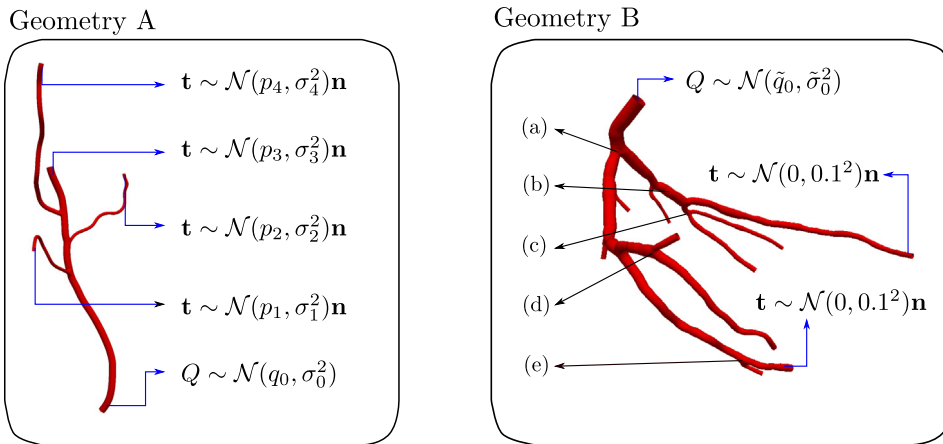


Fig. 9. Probabilistic data for complex geometries. (a) Isolated section from the intracranial system retrieved from the ADAN model. Uncertainties are imposed at the inlet and also at the four outlets with 10% of variability on each boundary ($\sigma_0 = 0.1q_0$, $\sigma_i = 0.1p_i$, $i = 1, \dots, 4$). (b) Left coronary arterial tree extracted from patient-specific medical images. Uncertainties are imposed at the inlet (10% of variability) and at two outlets (variability of 0.1). The other boundaries are outlets with deterministic zero traction.

a measure of the dispersion of the output distribution that is independent from the variable's unit measure. More specifically, the higher the CV, the higher the QoI variability.

The first geometry (left panel of Fig. 9) is a small structure composed by 7 branches, 3 bifurcations, and 5 global boundaries. This structure is decomposed in three subnetworks — one per bifurcation. Uncertainties are introduced at each global boundary, with a level of variability of 10%. Specifically, the inlet flow is equal to $q_0 = 0.25 \text{ cm}^3/\text{s}$ and, on the outlets, the normal component of the traction is defined by the values $p_1 = 1800 \text{ dyn/cm}^2$, $p_2 = 1500 \text{ dyn/cm}^2$, $p_3 = 1600 \text{ dyn/cm}^2$ and $p_4 = 2000 \text{ dyn/cm}^2$. These values represent the pressure difference with the diastolic pressure taken as reference value and are in agreement with physiological values. Notice that the number of uncertainty sources increases the dimension of the PCE space ($d = 5$) and also the number of samples needed for the convergence ($Q = 81$), in comparison with the simpler examples addressed in the scalability test section.

In Fig. 10 the pressure CV in the whole geometry as well as the detailed view of the WSS statistical moments on the three junctions are presented. As expected, the pressure CV is higher in proximity of the sources of the uncertainty (inlets and outlets), and lower in the interior of the domain, with a maximum value of the same order of the input variability percentage (10%). The pattern of the WSS spatial distribution in proximity of the bifurcations is

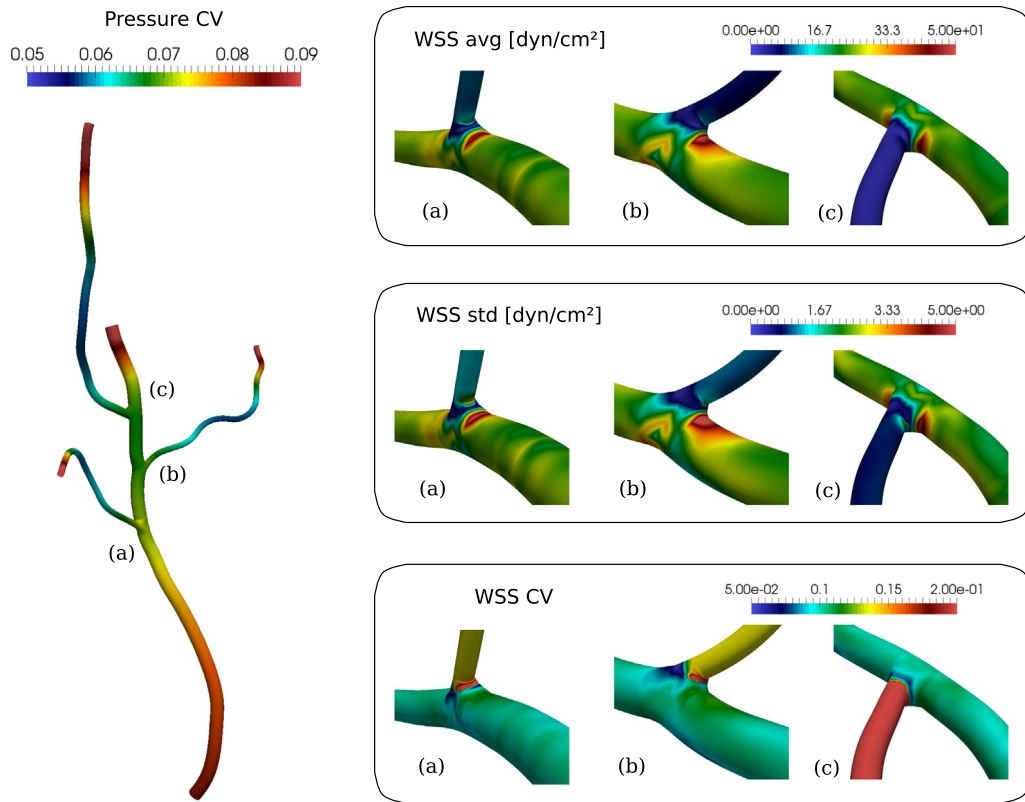


Fig. 10. Geometry A. Left: Global view of the coefficient of variation of the pressure computed via NetUQ-TEPEM strategy. Right: Quantities of interest related to the WSS at three selected regions.

similar in mean value and standard deviation. However, in this case the CV is much higher in the smaller vessels, indicating that a low mean WSS is typically characterized by a higher variability. This aspect rises interesting challenges, as it is well known that the regions more prone to develop pathology are generally associated with low (and oscillating) values of the WSS.

For the left coronary arterial tree (Geometry B), we consider three sources of uncertainty. On the inlet, a flow value of $\tilde{q}_0 = 0.5 \text{ cm}^3/\text{s}$ with variability of 10% is considered. Two outlets are selected, as seen in the right panel of Fig. 9, to consider a stochastic traction with zero mean value and a variation of 0.1. Over the other outlets, zero traction is considered. The geometrical complexity of this case is higher in comparison with the former cases. Therefore, a higher spatial variability of the QoI is expected. On this larger structure, the proposed strategy allows us to obtain a detailed spatial description of the statistical moments and, for a better visualization, five regions were selected to present the average, standard deviation and CV of the WSS field (see Fig. 11). Similarly to geometry A, the average and standard deviation spatial patterns are quite similar (but with different order of magnitude), and with a maximum value for the average of the same order of the imposed variability (10%). In the same figure we can see that the higher values of the CV are localized near the junctions (specially at the tip of each bifurcation), confirming the need of further research to gain insight about a possible relation between this (or other) statistical index with regions prone to develop cardiovascular diseases.

A detailed comparison of the errors summarized in Table 2 confirms the accuracy of the NetUQ method on such complex 3D geometries, even when using a reduced-order model such as TEPEM for sampling. As explained in Section 5.2, for small networks such as those considered here for realistic geometries, NetUQ is typically outperformed by the monolithic approach, which is confirmed by the computational times reported in Table 2. Nonetheless, based on the speedup results obtained in the scalability studies, we expect a similar speedup also for realistic networks at larger scales, which will be subject of future investigations. Moreover, while the monolithic computational time is more affected by the stochastic dimension, the NetUQ performance is more impacted by

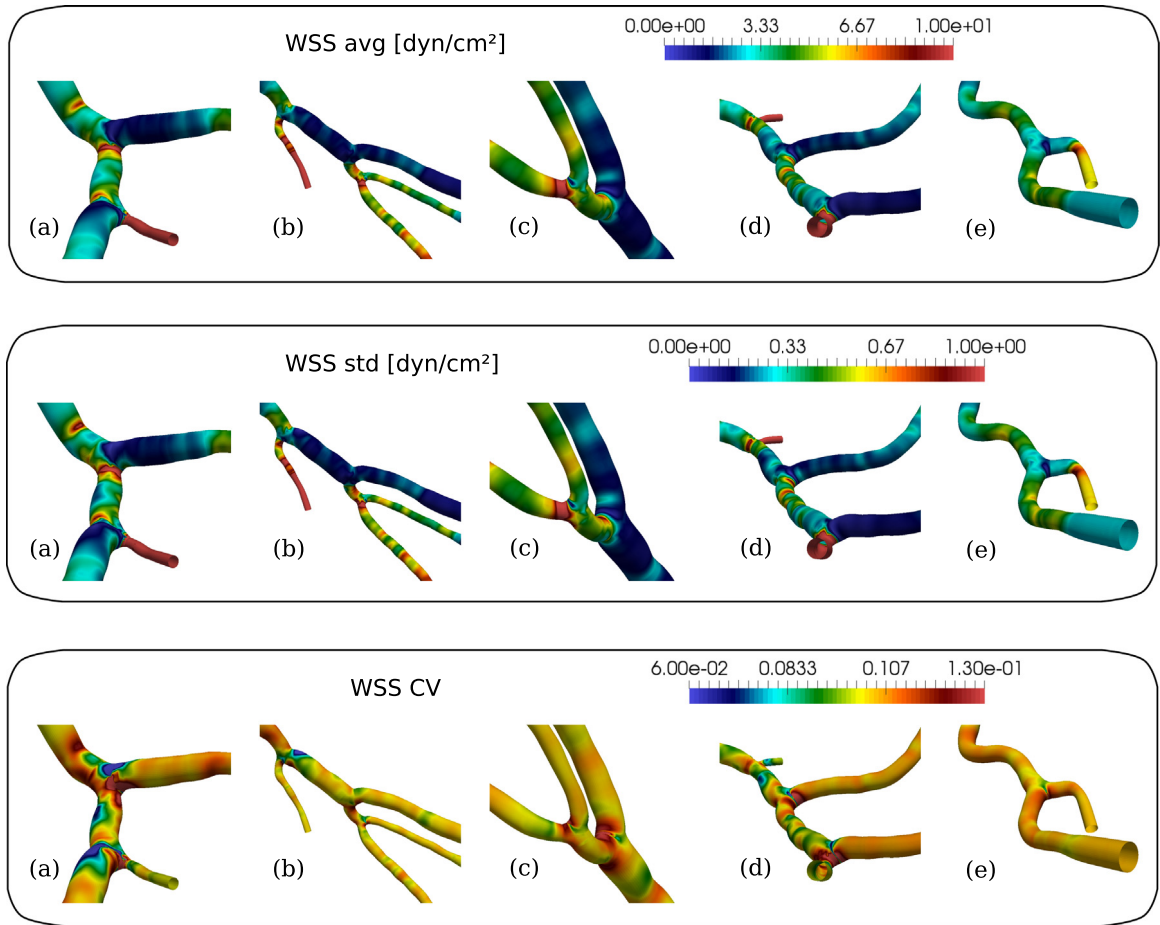


Fig. 11. Geometry B. WSS average (top), standard deviation (std) and coefficient of variation (CV) at five selected regions.

Table 2

Number of network components (DD) and number of pipe-elements (Pipe) for Geometries (Geo.) A and B; Dimension (d) and number of samples (Q) of the stochastic space; Relative error in the average and standard deviation for the pressure and wall shear stress between the NetUQ–TEPEM solution and the monolithic solution; Parallel computational time for NetUQ and monolithic (Mono.) approach.

Geo.	Components		PCE space		Pressure error		WSS error		Time	
	DD	Pipe	d	Q	Avg	Std	Avg	Std	NetUQ	Mono.
A	3	296	5	81	$8.84 \cdot 10^{-5}$	$5.87 \cdot 10^{-4}$	$1.26 \cdot 10^{-4}$	$1.06 \cdot 10^{-3}$	8 h 39'	3 h 36'
B	9	719	3	37	$2.71 \cdot 10^{-4}$	$1.74 \cdot 10^{-3}$	$2.75 \cdot 10^{-4}$	$2.15 \cdot 10^{-3}$	13 h 33'	3 h 06'

the network size. In fact, the monolithic computational times for Geometry A and Geometry B are comparable, in spite of different network sizes, since the stochastic space dimension d (and, consequently, the number of samples Q) is larger for the smaller network, resulting in a heavier local computational load (Geometry A: $n = 3$, $d = 5$, $Q = 81$ vs. Geometry B: $n = 9$, $d = 3$, $Q = 37$). On the other hand, with a NetUQ approach a larger network size (Geometry B) requires more iterations to convergence, resulting in a higher computational time.

6. Final remarks

In this paper we proposed a new approach to perform uncertainty quantification in the cardiovascular network that is computationally affordable and reliable. This is achieved by combining the NetUQ approach [38], which propagates uncertainties in large-scale networks in a scalable manner, with the TEPEM method for the steady

Navier–Stokes equations [25], which allows the hierarchical modeling of 3D fluid flow dynamics in a fraction of the time employed by conventional FEM strategies. The numerical results show that (i) weak scalability can be achieved for non-linear vector problems in realistic 3D domains; (ii) *educated* reduced models like TEPEM enable rapid and agile subsystem sampling in 3D domains, being computationally more affordable than full high-fidelity models, yet reasonably accurate; (iii) it is currently feasible to provide statistical information about quantities of clinical interest, such as the highly non-uniform pattern of the wall shear stress, which are completely out of reach for the traditional 1D models; (iv) stochastic simulations in patient-specific geometries achieve a good trade-off between computational affordability and accuracy, compared to the traditional models used in the literature.

There are several limitations in the present study. We solved the steady Navier–Stokes equations, as they can be interpreted as an approximation of time averaged quantities, quite often the main interest of clinicians, under periodic regimes. Another major limitation is that we considered rigid geometries, while 1D models intrinsically include fluid–structure interaction (FSI). As the wavelength of the pressure wave propagation is contained in the size of the network included in the computations, fluid–structure interaction is expected to have a major impact on the solution. In this respect, the present work has to be considered an intermediate step toward a complete FSI UQ analysis of the circulation. We are currently working on including FSI into TEPEM, which is all but a trivial task. The results presented here allow to argue that the combination of TEPEM and NetUQ could be a reliable and efficient approach for UQ in cardiovascular simulations even with patient-specific deformable vessels. This will be validated in future works.

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